Calibrated Prompt Uncertainty in Zero-Shot Segmentation: Full-Validation Evidence from Lightweight Methods and SAM

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Abstract

Promptable segmentation systems are often evaluated with clean points and boxes, although real prompts are uncertain and the two channels do not admit directly comparable perturbation scales. We present a pre-specified evaluation on all 1,449 PASCAL VOC 2012 validation images. A disjoint, class-balanced 100-image subset of VOC train is used only to calibrate point displacement and box perturbation to matched observable quality targets: point hit rate and box IoU. We compare Center Color, GrabCut, a transparent robust-superpixel method, three component ablations, and SAM ViT-B. Three primary hypotheses are tested with paired bootstrap intervals, sign-flip tests, and Holm correction. Robust Superpixel improves over Center Color by 0.0640 IoU (95% CI [0.0585, 0.0696]); SAM score selection improves over a single noisy prompt by 0.0193 [0.0135, 0.0251]; and matched-quality point-noise IoU exceeds box-noise IoU by 0.1069 [0.0997, 0.1138]. All effects survive Holm correction. Strong baselines and ablations qualify the contribution: GrabCut is stronger than the proposed lightweight method, the color seed and spatial prior are useful, and multi-box consensus has essentially zero average benefit. The result is a reproducible account of prompt-channel failure and evidence strength rather than a state-of-the-art claim.

1 Introduction

Interactive segmentation converts sparse user input into an object mask. Extreme points, iterative clicks, boxes, and foundation-model prompts have progressively reduced the inference-time supervision requirement [4, 5, 6, 7]. Segment Anything (SAM) established a general point-and-box interface with strong zero-shot transfer [7].

Prompt quality is nevertheless structured. A displaced point may remain inside an object, whereas a shifted box changes localization and spatial support. Equal numeric noise scales do not imply equal prompt difficulty. Prior work compares boxes, centroids, and random points for SAM [8], adapts with perturbed prompts [9], estimates SAM uncertainty [10], and measures the gap between simulated and human interactions [11]. We therefore measure the realized quality of each channel instead of assuming equal perturbation strength.

This study asks three confirmatory questions: whether a transparent robust-superpixel pipeline improves over a minimal color method; whether SAM's score selects better candidates than a single noisy prompt; and whether SAM remains more sensitive to box noise after point and box quality are approximately matched. The contributions are a hash-pinned train/validation protocol, observable-quality calibration, pre-specified paired inference with family-wise correction, strong CPU and SAM baselines, component ablations, and a data-free metric artifact.

2 Related Work

Interactive segmentation. GrabCut combines graph cuts and iterative appearance models [2]. DEXTR conditions segmentation on extreme points [4]; RITM and SimpleClick improve iterative click-based segmentation across datasets [5, 6]. Human studies show that simulated interactions do not automatically reproduce annotation behavior [11].

Promptable foundation models. SAM supports point, box, and mask prompts [7]. Variational prompting evaluations report that boxes can be stronger than centroid or random points [8]. PP-SAM studies perturbed-prompt adaptation [9]; UncertainSAM formalizes uncertainty estimation [10]; recent work also optimizes point prompts using learned agents [12]. These results establish prompt robustness as an active topic. Our narrower contribution is matched-quality full-validation evidence with pre-specified paired inference.

Transparent baselines. SLIC provides efficient, spatially regular Lab-space primitives [3]. We use SLIC as a representation rather than claim it as a new learned method. Comparison with GrabCut and explicit ablations prevents method complexity from being mistaken for superiority.

3 Methods

3.1 Task and prompts

For each VOC image, the largest connected foreground component across semantic labels 1–20 defines one binary target. The clean box is its tight rectangle; the clean point maximizes the Euclidean distance transform inside the target. These prompts use ground-truth geometry and define a controlled benchmark, not a model of unaided human annotation. No method is trained or fine-tuned on VOC validation.

3.2 CPU methods

Center Color estimates an RGB prototype around the point, thresholds color distance within the box, and applies morphology and point-connected component selection. Robust Superpixel computes SLIC labels and mean Lab/spatial features, estimates point-induced foreground and box-induced background prototypes, combines relative prototype distance with a spatial prior, unions the result with Center Color, and applies the same cleanup. Ablations remove the color seed, spatial prior, or three-box consensus.

GrabCut uses only the prompt: outside-box pixels initialize background, inside-box pixels probable foreground, and a point disk sure foreground. Five iterations are used. Eight OpenCV initialization failures are retained and scored as zero.

3.3 SAM and candidate selection

SAM ViT-B is evaluated with point-only, box-only, and point-plus-box prompts. Under noisy point-plus-box input, five extra prompts are sampled locally around the observed prompt without access to the clean prompt. We compare the single noisy prediction, maximum SAM score, mask-IoU consistency medoid, majority vote, and Oracle best-of-six. Oracle uses target IoU and is a descriptive upper bound only.

4 Frozen Protocol

4.1 Data separation

The tuning split contains 100 VOC train images: five largest-component targets per class, sampled with seed 20260719. The confirmatory split contains all 1,449 VOC validation rows. The prior 30-image study is exploratory and excluded from confirmation. Data manifests contain identifiers and geometry but no image or mask payloads; manifests, sources, algorithm files, SAM source, and the ViT-B checkpoint are SHA-256 pinned.

4.2 Prompt calibration

On tuning data only, independent deterministic grid searches match point hit rate and box IoU to the same quality targets using 20 trials per target.

Table 1: Tuning-only noise calibration.

Severity	Target	Point scale	Hit rate	Box scale	Box IoU
Mild	0.85	0.1675	0.8480	0.0400	0.8538
Moderate	0.65	0.2725	0.6495	0.1150	0.6509

On validation, moderate point hit rate is 0.6391 and box IoU is 0.6514. This difference is reported without post-hoc recalibration.

4.3 Hypotheses and inference

The primary metric is sample-macro IoU. Five SAM trials are averaged within each sample. H1 states Robust Superpixel exceeds Center Color. H2 states SAM score selection exceeds a single moderate noisy prompt. H3 states matched-quality box noise causes a larger loss than point noise, equivalently point-noise IoU minus box-noise IoU is positive.

Each paired effect uses a 20,000-replicate bootstrap interval and a 50,000-permutation sign-flip test. H1–H3 are corrected together by Holm's method at family-wise $\alpha = 0.05$. Dice, resource use, strata, alternative selectors, and Oracle are secondary or descriptive.

5 Results

5.1 CPU baselines and ablations

Table 2: Complete VOC validation CPU results. Failures are scored as zero.

Method	IoU	Dice	Median ms	Failures
Center Color	0.5404	0.6753	13.1	0
GrabCut point+box	**0.6859**	**0.7751**	416.8	8
Robust Superpixel	0.6044	0.7345	199.8	0
no color seed	0.4633	0.5977	187.0	0
no spatial prior	0.5827	0.7152	197.6	0
single box	0.6043	0.7344	188.5	0

H1 is supported: Robust Superpixel improves over Center Color by 0.0640 IoU, 95% CI [0.0585, 0.0696], Holm $p = 0.000060$. GrabCut is stronger. Removing the color seed costs 0.1411 IoU and removing the spatial prior costs 0.0217. Removing multi-box consensus changes mean IoU by only -0.00004, contradicting the exploratory intuition that box voting was important.

5.2 SAM modality

Table 3: Clean SAM ViT-B prompt modalities.

Prompt	IoU	Dice
Point only	0.5389	0.6235
Box only	0.8479	0.9058
Point+box	**0.8491**	**0.9087**

Point+box is only 0.0012 IoU above box-only. The full-validation result refines the exploratory claim: SAM is box-dominated, but box-only is not stronger in aggregate.

5.3 Primary hypotheses

Table 4: Pre-specified paired IoU effects with Holm correction.

Comparison	Mean Δ	95% CI	Holm p
H1: Robust Superpixel − Center Color	0.0640	[0.0585, 0.0696]	0.000060
H2: score selection − single noisy	0.0193	[0.0135, 0.0251]	0.000060
H3: point-noise IoU − box-noise IoU	0.1069	[0.0997, 0.1138]	0.000060

Moderate point noise yields 0.8194 IoU and matched-quality box noise 0.7125, supporting H3. Under joint noise, a single prompt reaches 0.6529, score selection 0.6723, consistency medoid 0.6758, vote consensus 0.6710, and Oracle

0.7822. H2 is statistically supported but modest. Consistency medoid was not a primary comparison and remains exploratory; Oracle is not deployable.

6 Discussion

Representative evaluation changes the original interpretation. The proposed CPU method is reliably better than its minimal baseline, but not GrabCut; its value lies in transparency and a speed–accuracy operating point. Box localization remains SAM's principal signal, consistent with prior prompting studies [8]; our added evidence is that channel sensitivity persists after realized prompt quality is approximately matched. SAM score selection becomes statistically reliable at full scale, although the improvement is only 0.0193 IoU and requires extra decoder calls.

Negative results are equally informative. Multi-box consensus adds complexity without aggregate benefit. The strongest CPU method is classical GrabCut. These findings prevent a polished engineering artifact from being presented as a stronger methodological contribution than the evidence supports.

7 Threats to Validity

Ground-truth-derived prompts are more controlled than real user input. Numeric matching of hit rate and box IoU does not make the metrics psychologically equivalent, and neither distribution is fitted to human errors. The benchmark selects one largest component per image rather than every instance. Only VOC 2012 and SAM ViT-B are tested; cross-dataset confirmation, SAM 2, larger checkpoints, and learned interactive baselines such as RITM or SimpleClick would improve external validity. GrabCut has eight explicit failures. The tuning split is class-balanced while validation has the natural distribution. Although source hashes prevent silent post-confirmatory changes, the hypotheses were motivated by exploratory results; an independent second dataset would offer stronger confirmation.

8 Conclusion

Under a frozen full-validation protocol, Robust Superpixel improves over Center Color but trails GrabCut. SAM depends strongly on box localization: matched-quality box noise produces a substantially larger loss than point noise. SAM score selection yields a small, Holm-corrected improvement over a single noisy prompt; Oracle remains only an upper bound. The primary contribution is a reproducible separation of engineering quality, prompt-channel sensitivity, and evidence strength rather than a state-of-the-art claim.

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